

Chapter – 2

Flipkart Online E-Commerce Database

2.0 Introduction

The entities required for the Flipkart Online E-Commerce Database are identified based on business requirements. The attributes of each entity, along with their primary keys, foreign keys, unique keys, and attribute types, are determined to ensure proper data organization. The relationships between the entities are also identified to represent the interactions within the system and establish a well-structured relational database design. Cardinality and participation constraints define how entities are associated. This serves as the foundation for the ER diagram, relational schema, and database implementation while ensuring data integrity, consistency, and minimal redundancy.

2.1 Entity: Customer

Attribute	Key Type
Customer_ID	Primary Key
First_Name	Not Null
Last_Name	-
Email	Unique Key
Phone	-
Address	Not Null

Entity: Seller

Attribute	Key Type
Seller_ID	Primary Key
Seller_Name	Not Null
Email	Unique Key
Phone	-

Entity: Category

Attribute	Key Type
Category_ID	Primary Key
Category_Name	Unique Key
Description	-

Entity: Product

Attribute	Key Type
Product_ID	Primary Key
Product_Name	Not Null
Seller_ID	Foreign Key
Category_ID	Foreign Key
Price	-
Brand	-
Stock	Not Null

Entity: Cart

Attribute	Key Type
Cart_ID	Primary Key
Customer_ID	Foreign Key
Grand_Total	-
Total_Items	-

Entity: Order

Attribute	Key Type
Order_ID	Primary Key
Customer_ID	Foreign Key
Order_Date	-
Order_Status	-
Total_Amount	-

Entity: Order_Item

Attribute	Key Type
OrderItem_ID	Primary Key
Order_ID	Foreign Key
Product_ID	Foreign Key
Quantity	-
Unit_Price	-

Entity: Payment

Attribute	Key Type
Payment_ID	Primary Key
Order_ID	Foreign Key
Payment_Method	-
Payment_Status	-
Payment_Date	-
Amount	-

Entity: Review

Attribute	Key Type
Review_ID	Primary Key
Customer_ID	Foreign Key
Product_ID	Foreign Key
Rating	-
Review_Text	-

Entity: Shipment

Attribute	Key Type
Shipment_ID	Primary Key
Order_ID	Foreign Key
Delivery_Address	-
Shipment_Status	-
Delivery_Date	-

2.2 Entity Relationships

Entity 1	Relation	Entity 2	Cardinality	Participation
Customer	Owns	Cart	One-to-One	Partial/Total
Customer	Places	Order	One-to-Many	Partial/Total
Customer	Writes	Review	One-to-Many	Partial/Total
Seller	Sells	Product	One-to-Many	Total/Total
Category	Categorizes	Product	One-to-Many	Total/Total
Cart	Contains	Order_Item	One-to-Many	Total/Total
Order	Includes	Order_Item	One-to-Many	Total/Total
Order	Has	Payment	One-to-One	Total/Total
Order	Has	Shipment	One-to-One	Total/Total
Product	Receives	Review	One-to-Many	Partial/Total

